

Building_Energy_Modeling_Documentation_Index

Building Energy Modeling — Documentation Index

Methodology, reference, and measure-definition documents for the EnergyPlus-based Building Energy Modeling (BEM) toolkit developed by the Resilient Habitat Lab.

Core methodology & reference

Document	What it covers
Methodology Overview	Building sources, simulation variants (DEFAULT / IAL / HPENV / EEM), and the key modelling assumptions (Buffalo CZ6A proxy, IAL-vs-HVAC COP, skip-when-better rules, airflow-network mode, high-rise geometric correction, downscaled prototypes, version routing).
Reference Standards	Published EUI benchmarks and source notes — DOE 2008, NRCAN 2018 medians, ASHRAE 90.1-2022, 2024 IECC. Sanity-check anchors, not interchangeable sources.
Canadian Baseline Transformation	Canadian baseline generation: NECB 2017 (commercial) + NBC 9.36 (single-family) transforms, internal-loads calibration, and compliance validation.
Climate Zone 6A Selection Rationale	Rationale for choosing Buffalo as the representative city for Montreal's Climate Zone 6A.
Photovoltaic Methodology	Plain-English companion for the PV generation columns in the master simulation reports; two-track (as-designed vs. upgraded) PV model.
Parallel IDF Preparation — Technical Reference	Technical reference for the parallel batch-preparation pipeline used ahead of large simulation runs.
Secondary School Layout Investigation	Case study on why a full-size secondary-school prototype does not fit a compact neighbourhood plot, and the resized-model resolution.

Energy Efficiency Measures (EEM) framework

Document	What it covers
Energy Efficiency Measures Framework	The active retrofit-measure ladder (DEFAULT / EEM1 / EEM2 / EEM3 / EEM4), per-measure detail, supporting literature, and result tables.

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